Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **SQL**-backed data platforms, practical **APIs**, and microservices in Python and **Java** on AWS and Azure, with schema design, unit testing, technical-debt reduction, and AI-driven workflow automation in **Agile** teams.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Data Retrieval, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed a multi-tenant PostgreSQL schema following **data warehouse** principles for 200+ users and 10K+ records, writing **SQL transformations** across PostgreSQL and SQL Server with indexing and query tuning that cut latency 35%.
- Built ETL-style integrations with FastAPI, Supabase, and SQL Server using incremental loads, reconciliation checks, and CI/CD-backed releases, processing 5K+ monthly jobs at 99%+ reliability alongside **AI workflow** orchestration for 1K+ jobs.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Designed Flask **REST APIs** and modular Blueprint abstractions for upload, rubric, and grading workflows backed by MySQL on AWS RDS, serving 150+ instructors and 10K+ daily requests in production.
- Automated grading with **AI-driven** workflows across GPT-4o, Claude, and Qwen-VL plus normalized usage logs tracking tokens and cost per call, cutting feedback turnaround 60% across concurrent batch workloads.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Remediated **technical debt** by refactoring API integrations into object-oriented Singleton and Factory modules, cutting debt 20% and improving response times 30% through pagination, field filtering, and caching.
- Wrote 20+ unit and integration suites with PyTest and Mockito in **Agile Scrum** sprints and redesigned AWS Glue and S3 ETL workflows, boosting pipeline efficiency 68% and reducing storage costs 15%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built Java Spring Boot **microservices** with RESTful APIs and WebSocket updates for a resource-management platform, defining service boundaries across 8 services and syncing 2M+ records across locations.
- Modeled Azure SQL schemas for customers, invoices, payments, and leads, and used **analytics** to guide fixes that cut resource wastage 42% and raised lead conversions 15% through Jenkins CI/CD releases.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Integrated multi-currency card, wallet, and PayPal **payment** flows with idempotent webhooks for refunds, disputes, and chargebacks, cutting manual refund handling about 70% while keeping booking APIs under 200ms.
- Built hourly Spark micro-batch **analytics** on Databricks over 50K+ booking and payment events, aggregating revenue, active users, and peak demand into dashboards for internal teams and client reporting.

Skills

Languages: Python, Java, SQL, T-SQL, TypeScript, JavaScript, Bash
Databases: PostgreSQL, SQL Server, Azure SQL, MySQL, Snowflake, Schema Design, Data Modeling, Query Optimization, Indexing
Backend & APIs: Spring Boot, FastAPI, Flask, REST APIs, Microservices, Object-Oriented Design, Design Patterns
Data & AI: ETL Pipelines, Spark, Databricks, AWS Glue, LLM Systems, Workflow Automation, Analytics
Cloud & Practices: AWS (S3, RDS), Azure, Docker, CI/CD, Jenkins, PyTest, Mockito, Unit Testing, Agile/Scrum, Code Reviews

Projects

Fund Product Data Service | *Java, Spring Boot, PostgreSQL, Snowflake, AWS, Mockito*

- Designed a test-driven **Java Spring Boot** API consolidating fund, share-class, fee, and performance data into a versioned relational model with **stored procedures** and Snowflake analytical views for downstream client experiences.

AI-Assisted Data Quality Workflow | *Python, FastAPI, PostgreSQL, LLM APIs, PyTest, AWS*

- Designed a Python workflow that runs SQL rule checks on incoming product data, routes failures to owners, and uses an **LLM** to summarize anomalies, with **test-driven** rule coverage in PyTest.

Chat2SQL Code | *Java, TypeScript, SQL Server, Snowflake, PostgreSQL, Docker*

- Extended an AI-assisted **SQL** workspace across 10+ engines including SQL Server, Snowflake, and PostgreSQL, adding schema synchronization and validated query execution while generated SQL reached 82% accuracy.